

Primers and Shortcuts: Testing Structural Preambles for LLM Graph Reasoning at Scale

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Abstract

Prepending a short *primer* of factual sentences about each node’s local structure — degree, clustering coefficient, a random-walk structural encoding, or connected-component membership — before a graph’s standard text encoding is an intuitively appealing way to ease graph-reasoning prompts for a large language model. Any apparent gain is illusory, however, if a solver can recover the answer directly from the primer text without ever consulting the graph, and it is confounded if the primer merely lengthens the prompt. We introduce a graph-blind shortcut solver as a content-only lower bound and a length-matched filler control as a length-only upper bound. Evaluated at $n = 40$ on qwen3-1.7b and qwen3-4b, with and without chain-of-thought — 67,200 generations across six GraphQA tasks, seven primer conditions and four edge densities — we find that no primer effect replicates across model scale. Several cells beat both *none* and *filler* significantly in one model size but *none* does so in both, and the largest consistent content effect — the *degree* primer on *node_degree*, which states the answer verbatim — reverses sign between 1.7B (+7.5 points) and 4B (−6.5). The filler is not inert: it costs the thinking arm up to 14.1 points on dense graphs and harms both sizes on *edge_existence*, while the largest apparent gains fall on tasks whose answer is effectively constant, where content-free text moves accuracy as much as a structural primer does. Chain-of-thought on an identical checkpoint is worth +29.2 points pooled, an order of magnitude more than any primer effect measured. One cell survives every control: clustering on *node_degree* for qwen3-1.7b, +3.8 points, replicated at +4.3 on an independent sample ($p = 0.0006$).

1 Introduction

Fatemi et al. (2023) demonstrate that a large language model’s accuracy on graph-reasoning ques-

tions depends substantially on how the graph is encoded as text, independent of the model’s underlying reasoning ability. Their study varies the encoding function itself — adjacency lists, incident lists, and a family of “themed” encodings that rename nodes as people, places, or characters — while holding the question and the underlying graph fixed.

This work poses a complementary question: rather than altering how edges are described, does prepending a short *primer* of derived, node-level facts before the graph encoding help a model answer questions about that graph? A primer lies outside the encoding vocabulary studied by Fatemi et al. (2023): it is additional, redundant information computed from the graph and expressed in plain sentences, a term this project introduces for it. The appeal is obvious — if a model miscounts a node’s neighbours in a dense incident list, telling it the degree outright ought to help — and so is the risk.

Two confounds make such an effect hard to read, and both are easy to overlook. If a primer condition helps only because it lets the model read the answer directly from the primer text rather than reason over the graph, that is not evidence of increased graph reasoning; it is evidence that the primer leaked the answer. And if a primer helps or hurts only because it lengthens or shortens the prompt, that is not evidence about its *content* either: Liu et al. (2024) and the needle-in-a-haystack protocol of Kamradt (2023) show that context position and length alone move retrieval accuracy, independent of what the added text says. A primer study that compares each condition only against a no-primer baseline measures the sum of three things — content, leakage and length — and reports it as one.

This paper makes two contributions aimed squarely at separating them. First, a single, deterministic primer renderer that produces seven conditions (Section 3.1) differing only in content, so that differences between conditions are inter-

pretable as content effects rather than formatting artifacts. Among those seven is a length-matched, information-free *filler* condition that bounds how much of any effect is attributable to prompt length alone. Second, a *shortcut solver* (Section 3.2) that answers each GraphQA task from rendered primer text alone, with no access to the graph, yielding a principled lower bound that any genuine primer benefit must exceed. We evaluate both within a density sweep of GraphQA tasks on two model sizes, qwen3-1.7b and qwen3-4b, each with and without chain-of-thought (Section 4).

The result is largely negative, and the controls are what make it informative rather than merely disappointing: of the effects that do reach significance, the reliable ones are answer retrieval, the large ones are length, and the headroom in which anything else could have been observed is mostly consumed by the decoding mode.

2 Related Work

Fatemi et al. (2023) introduce GraphQA, a benchmark of graph reasoning tasks (node degree, node count, edge count, edge existence, connected nodes, cycle check, and disconnected nodes) paired with a family of graph-to-text encoding functions, and show that encoding choice, task, and the underlying graph generator all materially affect LLM accuracy. This project adopts their task definitions and their finding that the graph generator matters as motivation for fixing graph size and edge density explicitly in our own sweep design (Section 4), rather than sampling graph structure incidentally.

We verified this directly before finalizing the design, and the result is worth recording because it closed off an experiment we had intended to run. Sweeping the generator itself (Erdős–Rényi (Erdős and Rényi, 1959), Barabási–Albert, random regular, Watts–Strogatz, and stochastic block models) at matched node and edge counts moved the majority-class baseline from 0.15 to 1.0 — a swing six times larger than any primer effect measured anywhere in this project, with random-regular graphs degenerate in the sense that every instance shares a single gold answer. A generator-family sweep was accordingly withdrawn: it confounds structural family with task difficulty rather than isolating a structural effect, and a primer effect measured across it would be uninterpretable for exactly the reason the *filler* control exists.

A broader literature probes LLM graph reason-

ing more generally. Wang et al. (2023) evaluate eight graph tasks of increasing complexity and report that LLMs are brittle to spurious correlations in graph and problem structure, the same concern that motivates our shortcut solver (Section 3.2), applied here to the primer channel rather than to the graph encoding. Guo et al. (2023) instead encode graphs through standard graph description languages (GML, GraphML) and find that LLMs struggle to exploit that structure directly, and Jin et al. (2024) survey the broader space of LLM-graph integration methods within which this project sits. Dziri et al. (2023) argue that transformers solve compositional tasks — multi-digit multiplication, dynamic programming — by linearized sub-graph matching rather than systematic multi-step reasoning, a finding that raises the same question a primer’s apparent benefit raises: is an observed gain evidence of reasoning, or evidence of a shorter pattern to match against? Geirhos et al. (2020) frame this concern as *shortcut learning* in general: a decision rule that performs well on the benchmark it was measured against without solving the intended problem. Our shortcut solver operationalizes precisely this concern for the primer channel, rather than for the model as a whole.

The RWSE primer condition (Section 3.1) adapts the random-walk structural encoding of Dwivedi et al. (2022), originally developed for graph transformers rather than for text prompts; we render its return probabilities as natural-language sentences rather than feeding them to an attention mechanism. The chain-of-thought variant of our sweep follows Wei et al. (2022), who show that eliciting step-by-step reasoning changes what a model can accomplish on structured tasks, consistent with the reasoning-mode effect we observe in Section 5.6, which dwarfs every primer effect measured. Our significance-testing methodology (Section 3.4) follows the general recommendation of Dror et al. (2018) to test the actual dependence structure of paired evaluation data rather than treat each item as independent.

3 Method

3.1 Primers

A primer is a short preamble of factual sentences that details a graph’s local or global structure, prepended to the standard text encoding and the task question. To ensure that observed differences reflect genuine content effects rather than format-

Condition	Chars	Rendered form (node 0)
none	0	—
components	39	(<i>one sentence, graph-level</i>)
degree	897	“has degree 5.”
clustering	1631	“has clustering coefficient 0.20.”
filler	1831	“is simply present within G .”
rwse	2951	“has return probability 0.27 after 2 steps and 0.04 after 3 steps.”
all	4697	(<i>the three above, conjoined</i>)

Table 1: The seven conditions at $n=40$: median rendered primer length in characters over 400 prompts, and the sentence each emits for a single node. Every node-level condition emits exactly one sentence per node, so the conditions differ in what the sentence asserts and in nothing else. `components` emits one sentence for the whole graph, which is why it is two orders of magnitude shorter than the rest — the one condition whose effect cannot be confounded with length.

ting artifacts, a single renderer produces every condition: all node-level facts consequently share identical sentence framing (e.g., “Node N has. . .”), and floating-point values are rounded and formatted identically across conditions. We evaluate seven distinct conditions:

- **None:** The baseline graph text encoding without a primer.
- **Components:** A single graph-level sentence stating the number of connected components.
- **Degree:** The exact degree reported for each node.
- **Clustering:** The clustering coefficient computed for each node.
- **RWSE:** The random-walk structural encoding (Dwivedi et al., 2022) return probabilities for each node.
- **All:** The degree, clustering, and RWSE statistics combined for every node.
- **Filler:** A content-free control matched in length to the factual primers (e.g., “Node N is simply present within the graph G .”), isolating the effect of primer content from the confound of merely lengthening the prompt.

Table 1 gives the rendered length of each condition at the graph size used throughout. Two features of it matter for interpretation. `components` is 39 characters against `all`’s 4,697, so it is effectively a length-free intervention and any effect

Task	none	comp.	clust.	rwse	degree	filler	all
node_count	.064	.076	1.00	1.00	1.00	1.00	1.00
edge_count	.018	.018	.148	.018	1.00	.018	1.00
node_degree	.082	.082	.082	.616	1.00	.082	1.00
connected_nodes	.082	.082	.082	.082	.208	.082	.352
edge_existence	.498	.498	.720	.642	.794	.498	.794
cycle_check	.832	1.00	.832	.832	.946	.832	.946

Table 2: Shortcut ceilings: accuracy of a solver that reads only the rendered primer and never the graph, over 500 graphs from the GraphQA generator. **Bold** marks cells the primer fully determines, on which no model result can be evidence of graph reasoning. Four of six tasks are decided this way for at least one primer. Note that a 1.00 here is what a Python program scores, not what a model scores: on `node_count` the ceiling is 1.00 and Fatemi et al. (2023) report 18.8% for PaLM 2 on the encoding GraphQA ships. A decided cell still answers a question, but a narrower one — whether the model uses a fact it was handed, not whether it reasoned about the graph.

it shows is content. And `filler`, at 1,831 characters, falls between `clustering` (1,631) and `rwse` (2,951): it brackets the informative primers rather than matching any single one exactly, so it bounds the length term rather than cancelling it.

3.2 Shortcut solvers

For a primer condition’s measured accuracy on a task to constitute evidence of graph reasoning, it must exceed what is recoverable from the primer text alone. We construct a shortcut solver that takes only rendered primer text as input — never the graph object, and never the full-precision statistics the primer rounds away — and answers each GraphQA task directly. It combines sixteen exact rules (for instance, deriving edge count algebraically from a `degree` primer, since the stated degrees sum to $2m$), one heuristic rule, eight fitted rules whose fitting and scoring graph sets are kept structurally disjoint so that accuracy is not inflated by evaluating on the fitting data, and an exact enumeration bound for small graphs.

The solver is *structurally* incapable of observing the graph: it takes rendered text as its only argument, so there is no graph to pass and no discipline to violate. Its accuracy on a given (primer condition, task) cell is therefore a model-independent bound. Any model result on that cell should be interpreted relative to this *shortcut ceiling* rather than to zero, since a primer condition that fully determines the answer (shortcut accuracy of 1.0) renders model accuracy on that cell uninformative about graph reasoning, however high it may be.

Table 2 gives the resulting ceilings. They sort the sweep rather than pruning it. Three of the sixteen exact rules *reconstruct* rather than compare: a stated degree sequence constrains which graphs are possible, often to exactly one, so a primer that states no adjacency at all can still give away whole neighbour lists. That is why `connected_nodes` under `all` is a 35.2% cell rather than the 8.2% one it first appears to be.

3.3 Prompt assembly

Each prompt concatenates the primer, the graph’s text encoding, and the task question. Prompts are rebuilt from a canonicalized, parsed representation of each graph rather than reused verbatim from the source dataset, so that node and edge ordering is reproducible and independent of the order in which the original question happened to list them. We use the `incident` encoding throughout, which emits one line per node; the published dataset ships the `adjacency` encoding in its `question` field, which is why prompts are regenerated rather than reused.

3.4 Significance testing

Every comparison reported here is paired: a primer condition is compared against `none` on the same graph, with the same question and gold answer, and pairs in which either arm reached the output-token cap are dropped from that comparison rather than scored as incorrect (Section 5.8 reports what the opposite convention would do). Because a graph contributes exactly one pair to a given (task, condition) comparison, the pairs entering a single test are independent and no clustering correction is required; we do not pool across tasks, where they would not be (Dror et al., 2018).

The primary test is the exact McNemar test (McNemar, 1947), Benjamini–Hochberg corrected across the six primer conditions within a task (Benjamini and Hochberg, 1995) — $q = 0.05$, six hypotheses per family, rather than one family of 144, since a primer’s effect on `node_count` and its effect on `cycle_check` are not alternative answers to one question. For every comparison we additionally report a paired permutation test (sign-flipping each pair’s observed difference, 20,000 resamples with the add-one correction of North et al.) and a paired bootstrap 95% confidence interval on the effect size (20,000 resamples, whole pairs resampled together; Appendix A). The permutation and McNemar p -values agree to within a few

thousandths on every cell, so the interval, not the second p -value, is what the extra computation buys.

3.5 Answer extraction and measurement validity

Turning free model text into a scored answer is the quiet source of measurement error in a study of this shape, and it is not symmetric across conditions: chain-of-thought responses are longer and messier than zero-shot ones, so a naive extractor makes the thinking arm look worse than it is — which is one of the comparisons we want to draw. Extraction is therefore written per task and tested against the shapes models actually emit, rather than being a single regular expression.

One failure is worth reporting in full, because it silently produced a headline-sized effect that does not exist. Our boolean extractor took the last yes/no token in the response. On `cycle_check`, `qwen3-4b` frequently answers correctly and then recites the definition: “*Yes, there is a cycle. A cycle is a path . . . with no repeated edges or nodes.*” The trailing “no” became the extracted answer. Correcting the extractor to ignore the phrase “no repeated” changed 225 of `qwen3-4b`’s 5,568 scored boolean rows, every one of them from wrong to right, and every one on `cycle_check`; no row moved in the other direction. Before the fix, `filler` scored 0.140 at $p=0.2$ and `none`, `all` and `rwse` scored 0.86, 0.64 and 0.51 at $p=0.5$, which reads as a dramatic condition-dependent collapse. After it, every condition scores 0.97 to 1.00 and nothing on that task is significant. All numbers in this paper use the corrected extractor. We note this not as a curiosity but as a pattern: an extraction bug that correlates with response style will manifest as a condition effect, and there is nothing in the resulting table that looks wrong.

4 Experimental Setup

Tasks. We evaluate on six GraphQA tasks from Fatemi et al. (2023): node count, edge count, node degree, connected nodes, edge existence, and cycle check.

Models. We compare `qwen3-1.7b` and `qwen3-4b` from the Qwen3 model family (Qwen Team, 2025), each evaluated in a non-thinking (greedy decoding, no chain-of-thought) and a thinking (chain-of-thought-enabled; Wei et al., 2022) variant, for four model arms in total. Both families are pinned explicitly rather than left at

their defaults, which are opposite: left alone, Qwen3 would reason in a hidden channel even in the nominally non-thinking arm, collapsing the two arms into one condition.

Sweep design. All four arms are evaluated on one shared set of prompts, so that every model receives byte-identical inputs. Graphs are Erdős–Rényi (Erdős and Rényi, 1959) with a fixed node count of 40 and edge density swept over four levels ($p \in \{0.10, 0.20, 0.35, 0.50\}$). Every combination of the seven primer conditions (Section 3.1) and six tasks is evaluated on 100 graphs per (density, task) cell, yielding 16,800 prompts per model arm and 67,200 generations in total. Generation is greedy, with a raised output budget (up to 8192 tokens), since at the densest level tasks such as edge count require enumerating several hundred edges in the answer.

Auxiliary sweeps. Two comparisons need more power or a wider density range than the main sweep provides, and use dedicated prompt sets built by the same renderer. The first fixes `node_degree` and 400 graphs per density level over the same four densities, for the clustering comparison of Section 5.7, and was run twice on independently drawn graph samples. The second extends the density range to $p \in \{0.10, \dots, 0.85\}$ on `node_degree` for the length and chain-of-thought comparisons of Sections 5.5 and 5.6, where the main sweep’s ceiling at $p \leq 0.50$ leaves too little dynamic range to measure either. Both are identified explicitly wherever their numbers appear.

Metrics. We use exact match for integer- and boolean-valued tasks and set-F1 for connected nodes. Every result is reported against both the `none` baseline and the shortcut ceiling for the corresponding (primer condition, task) cell (Table 2). Generations that reach the output-token cap are excluded from scoring rather than scored as incorrect, and are reported separately as a per-cell drop count.

Note that we report `connected_nodes` on exact match in the main table even though F1 is its primary metric. Its F1 sits at 96–100% for every condition in every arm, which conceals differences that exact match resolves; and a McNemar test on F1 binarized as “> 0” is degenerate, since a partial-credit metric is almost never exactly zero, so it reports 0 to 1 discordant pairs everywhere and would be misread as “no effect”. The F1 figures are in Appendix B.

Task	Modal answer	0.10	0.20	0.35	0.50
<code>node_count</code>	“40”	1.000	1.000	1.000	1.000
<code>cycle_check</code>	“Yes”	1.000	1.000	1.000	1.000
<code>edge_existence</code>	“No”	0.900	0.820	0.690	0.520
<code>node_degree</code>	integer	0.210	0.170	0.170	0.180
<code>edge_count</code>	integer	0.083	0.065	0.046	0.056
<code>connected_nodes</code>	node set	0.019	0.010	0.010	0.010

Table 3: Blind baselines: accuracy of always returning the modal gold answer, by density. This is the floor a model must clear before any primer effect on that task means anything, and it is the reason `node_count` and `cycle_check` carry no information at $n=40$ — their gold answer is constant, so a model can be perfect without reading the graph. Only `node_degree`, `edge_count` and `connected_nodes` have real headroom at every density.

Reproducibility. All generation is greedy (no sampling), so a run is deterministic given a checkpoint and a prompt file; the prompt files are written to disk before any generation and shared byte-identically across arms. Thinking and non-thinking arms are the same checkpoints with the model’s native reasoning channel pinned on or off, rather than a prompt-level chain-of-thought instruction, so the two arms differ in decoding mode alone. Code, prompt sets and raw generations are released with the paper.

Known validity caveats. At this graph size and density range, several cells are known in advance to be shortcut-dominated rather than genuine tests of graph reasoning: any primer with one sentence per node allows node count to be recovered by counting sentences; cycle check’s gold answer is “yes” for nearly every graph past the sparsest density level; and the `degree` primer states the node-degree answer verbatim. The one cell already established as free of this contamination at this graph size is `node degree` under `{none, components, clustering, filler}`. We report and interpret results with this caveat in mind rather than omitting the contaminated cells, since a null or negative result on an uncontaminated cell remains informative in its own right.

Table 3 makes the last caveat quantitative, and it is important enough to state plainly: at $n=40$ the constant answer “40” scores 100% on `node_count` and the constant answer “Yes” scores 100% on `cycle_check`. Neither task measures graph reasoning in this design. We report them because they turn out to measure something else worth knowing (Section 5.3), not because they bear on the primer hypothesis.

Task	1.7B		4B	
	plain	think	plain	think
edge_count	22.4	81.0	0.6	62.3
cycle_check	6.1	6.5	1.0	1.9
all others (max)	0.8	2.5	0.1	3.5
Overall	4.8	15.4	0.3	11.7

Table 4: Truncation rate (% of generations reaching the 8192-token output cap), by task and model arm.

<i>p</i>	none	comp.	clust.	rwse	degree	filler	all
<i>qwen3-1.7b (plain)</i>							
0.10	31	20	10	10	10	15	26
0.20	9	20	3	9	6	9	9
0.35	50	33	26	21	2	40	33
0.50	29	36	33	32	20	41	44
<i>qwen3-4b-think</i>							
0.10	60	27	20	31	64	40	50
0.20	66	34	47	48	52	50	47
0.35	92	65	51	91	59	77	63
0.50	100	92	96	99	60	99	63

Table 5: edge_count: generations dropped to truncation, out of 100 per cell. The censoring is severe and *condition-dependent*, which is worse than severe and uniform: at $p=0.5$, none loses every row for qwen3-4b-think while degree loses 60, so the two arms of that comparison are scored on different graphs. The qwen3-1.7b-think arm (not shown) has cells at 100% from $p=0.35$ upward.

5 Results

All four model arms completed the full $n=40$ density sweep: 16,800 scored generations per arm, one per (task, primer, density, graph) cell. Unless stated otherwise, every reported Δ is a *paired* difference against none on instances for which neither arm reached the output-token cap, tested as described in Section 3.4. Complete per-density tables for every (arm, task) pair are in Appendix B, and effect sizes with confidence intervals for all 144 cells in Appendix A.

5.1 Truncation must be read before accuracy

Table 4 reports the fraction of generations that reach the output-token cap. The rate is neither uniform across tasks nor across arms, and it is not ignorable: on edge_count the thinking arms truncate on 62–81% of instances, with individual (density, primer) cells at 100%. Because truncated generations are dropped rather than scored as incorrect, the surviving sample in those cells is a survivorship-biased subset of the sparsest graphs, and the apparently high none accuracy there (86.3% at 1.7B,

97.6% at 4B, against 2% accuracy in the corresponding untruncated plain arms) measures which graphs finished, not which were solved.

Table 5 shows why this cannot be repaired by a footnote. The drop rate varies by primer within a density, so the pairs surviving a degree-vs-none comparison are not a random subset of the pairs that entered it. We therefore report edge_count under chain-of-thought but decline to interpret it; several of its comparisons rest on fewer than 60 surviving pairs, and one of the largest apparent effects in the entire sweep (−61.5 points for degree on qwen3-1.7b-think) rests on 13.

5.2 The full density sweep

Table 6 reports the full sweep and Figure 1 the density structure underneath one column of it. Four observations organize them.

No primer effect replicates across model scale.

Several (primer, task) cells significantly exceed both none and filler in one plain arm; none does so in both. The strongest candidate is the degree primer — and the all primer that contains it — on node_degree, where the primer states the requested quantity verbatim: it is worth +7.5 [+3.0, +12.0] and +10.2 [+5.2, +15.2] points to qwen3-1.7b and +11.7 and +11.6 to qwen3-1.7b-think, but −6.5 [−9.2, −4.0] and −9.2 [−12.2, −6.2] to qwen3-4b; the 1.7B and 4B intervals do not overlap in either primer. A primer that *contains the answer* makes the larger model worse. The failures are not random: 26 of qwen3-4b’s 29 degree losses are off by one to three, and its mean absolute error rises from 0.015 under none to 0.138 under degree, consistent with the primer competing with, rather than replacing, the model’s own enumeration of the incident-list encoding.

Answer retrieval is partial even where it is complete in principle.

The shortcut solver attains 1.00 on node_degree under degree and all, and on edge_count under the same two primers (Table 2). No model arm approaches that ceiling. The largest edge_count gain anywhere in the sweep is qwen3-4b under degree at +27.7 points, which takes the arm from 2.0% to 29.7% and still leaves roughly 70% of instances wrong against a bar of 100%. A primer that fully determines the answer is therefore not a test of graph reasoning, but neither is it automatically exploited; what it measures is whether the model uses a fact it

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
qwen3-1.7b	connected_nodes	72.5	+0.8	+1.2	+1.0	+3.5	-6.2	+2.0
	cycle_check	99.7	-0.6	+0.3	-7.5	-12.6	+0.3	-1.1
	edge_count	1.8	-1.4	-0.4	-0.8	+4.0	-0.4	+2.4
	edge_existence	70.5	-8.2	+3.8	+5.3	+4.0	-15.8	+11.5
	node_count	31.5	+21.2	+67.2	+2.0	+6.8	+39.8	+47.0
	node_degree	60.5	+1.8	+3.0	+4.0	+7.5	-0.5	+10.2
qwen3-1.7b-think	connected_nodes	79.7	+2.8	+1.5	-0.3	+7.4	+2.0	+6.1
	cycle_check	100.0	0.0	0.0	0.0	-0.7	0.0	0.0
	edge_count [†]	86.3	+7.4	-8.7	-4.0	-61.5	+3.4	-8.3
	edge_existence	99.2	-0.8	-1.0	-1.3	-2.3	-0.8	-2.0
	node_count	98.6	+0.9	+1.4	+1.4	+0.8	+1.4	+1.4
	node_degree	76.8	-1.0	+1.0	+2.5	+11.7	-3.5	+11.6
qwen3-4b	connected_nodes	86.0	+9.2	+1.0	0.0	-3.0	-12.5	-2.8
	cycle_check	99.7	-1.3	+0.3	+0.3	-1.5	-0.3	-0.5
	edge_count	2.0	-1.3	-1.3	-1.5	+27.7	+1.2	+3.8
	edge_existence	90.5	-2.8	+4.0	-10.0	-4.3	-7.0	-9.8
	node_count	90.2	+9.0	+8.8	+9.2	+8.0	+9.2	+9.8
	node_degree	99.2	+0.2	-0.5	-2.2	-6.5	-0.5	-9.2
qwen3-4b-think	connected_nodes	98.7	+0.5	0.0	0.0	0.0	+0.8	-0.5
	cycle_check	100.0	0.0	0.0	0.0	-0.8	0.0	0.0
	edge_count [†]	97.6	+1.7	+1.7	0.0	-23.5	-2.0	-41.5
	edge_existence	100.0	0.0	0.0	0.0	0.0	0.0	0.0
	node_count	99.5	0.0	+0.5	+0.5	+0.5	+0.5	+0.5
	node_degree	99.7	-0.3	-0.3	-2.1	0.0	0.0	-0.8

Table 6: Main sweep, $n=40$, pooled over densities $p \in \{0.10, 0.20, 0.35, 0.50\}$, 100 graphs per (task, primer, density) cell. The `none` column is absolute accuracy (%); all other columns are paired differences against `none` in percentage points. Exact match throughout, including `connected_nodes` (see Section 4). Truncated generations are excluded pairwise, giving $n \in [376, 400]$ paired instances per cell except where marked. **Bold** indicates significance under an exact McNemar test after Benjamini–Hochberg correction across the six primers within a task ($q = 0.05$). [†]Censored beyond interpretation (Table 5); $n \in [13, 60]$ surviving pairs.

was handed, and the answer is that it largely does not.

The shortcut ceilings do not transfer to this graph size. They were fit on GraphQA’s native generator, whose graphs have 5–19 nodes. At the fixed $n=40$ used here, the blind baselines of Table 3 dominate them on two tasks: the constant answer “40” scores 100% on `node_count` and the constant answer “yes” scores 100% on `cycle_check`, so the tabulated ceilings understate the trivial baseline for those tasks by a wide margin. Only ceilings backed by exact theorems — `degree` and `all` on `node_degree` and `edge_count` — carry over unchanged, and we read ceiling-adjusted effects only on those.

The two controls disagree about which cells are interesting. This is worth stating because it is the point of having both. `components` is nearly length-free (39 characters), so its effects are content by construction; it is also inert almost everywhere, with the exception of `node_count` and `connected_nodes` at 4B. `all` is the longest primer and shows the largest and least consistent

swings, including +11.5 and -9.8 on the same task in different model sizes. Reading either column alone would produce a different account of the study.

5.3 Trivially-answerable tasks respond to text, not to structure

The largest effects in Table 6 appear on `node_count`, where the gold answer is the constant 40 (Table 3) and the question is therefore not a graph-reasoning question at all. At 1.7B every condition improves over `none` (31.5%) by between +2.0 and +67.2 points, and the ordering follows no property of primer content: clustering +67.2, `all` +47.0, `filler` +39.8, `components` +21.2, `degree` +6.8, `rwse` +2.0 (n.s.). At 4B the six conditions are statistically indistinguishable from one another, all improving `none` by +8.0 to +9.8 points. In both arms the content-free `filler` control is among the strongest conditions.

The mechanism is mundane and worth naming. The dominant `none` failure is the answer “39”, an off-by-one induced by the encoder’s “nodes 0, ..., 39” listing; the model reports the largest

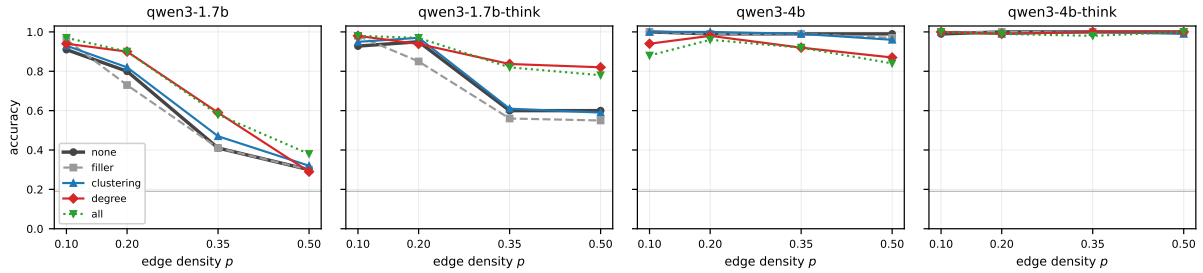


Figure 1: `node_degree` accuracy against edge density, all four arms, five conditions. The grey line is the blind baseline (Table 3). The panels show why a single pooled number for “the primer effect” is not well defined: `qwen3-1.7b` has a steep difficulty gradient and admits a positive effect, `qwen3-4b` is saturated under `none` at every density so the only movement available is downward, and both thinking arms are at ceiling throughout. `degree` and `all` — the two conditions that state the answer — are the only ones that separate from `none`, upward at 1.7B and downward at 4B.

Condition	density p			
	0.10	0.20	0.35	0.50
<i>qwen3-1.7b</i>				
none	0.910	0.800	0.410	0.300
clustering	0.930	0.820	0.470	0.320
degree	0.940	0.900	0.590	0.290
filler	0.960	0.730	0.410	0.300
all	0.970	0.900	0.580	0.380
<i>qwen3-4b</i>				
none	1.000	0.990	0.990	0.990
clustering	1.000	1.000	0.990	0.960
degree	0.940	0.980	0.920	0.870
filler	1.000	0.990	0.990	0.970
all	0.880	0.960	0.920	0.840

Table 7: `node_degree` accuracy by density, plain arms, selected conditions; the numbers behind the first two panels of Figure 1.

index rather than the count. Any interposed text appears to disrupt that reading, and a primer that emits one sentence per node additionally makes the count recoverable by counting sentences. We accordingly retract the reading of this cell as a structural primer effect — an earlier version of this work reported components on `node_count` as a content gain that replicated across arms, and it does replicate, but so does `filler`.

5.4 Density and the ceiling problem

Table 7 shows why the sign reversal above is not a contradiction but a ceiling artifact. `qwen3-4b` answers `node_degree` correctly on 99% of instances under `none` at every density in the sweep; the task supplies no headroom in which a primer could help, and the only measurable primer effect is the cost of the added tokens. `qwen3-1.7b` degrades from 0.910 to 0.300 over the same density

range and does admit a positive effect.

This generalizes beyond the one task. A cell that is saturated for one model and taxing for another cannot yield a model-independent primer effect, and an estimate pooled over the two arms averages a ceiling against a floor. Of the 24 (arm, task) cells in Table 6, 10 sit at or above 98% under `none` and therefore admit no upward effect at all; 2 are censored past interpretability (Section 5.1); and 2 more have effectively constant gold answers. The design question a primer study has to answer before measuring anything is which cells are left.

5.5 Prompt length is not a neutral covariate

The `filler` condition is not inert. It significantly harms `qwen3-1.7b` on `edge_existence` (-15.8 $[-20.0, -11.8]$) and `connected_nodes` (-6.2 $[-10.2, -2.2]$), and `qwen3-4b` on `connected_nodes` (-12.5 $[-17.0, -8.0]$) and `edge_existence` (-7.0 $[-10.8, -3.5]$), while significantly *helping* both arms on the trivially-answerable `node_count`. Since it carries no structural information by construction, any effect it registers is attributable to length and position alone (Liu et al., 2024; Kamradt, 2023), and effects of comparable magnitude measured against `none` for the informative primers cannot be credited to content without this control.

Table 8 bounds the confound directly. The effect is not constant: `filler` is harmless, even mildly helpful, at $p \leq 0.20$, and turns sharply negative from $p = 0.50$ upward. That shape matters for reading the main sweep, whose densest level is 0.50: the length penalty there is real but near the bottom of its range, so Table 6 understates how much of a

p	n	Δ	p -value
0.10	390	+2.3	0.078
0.20	397	+2.5	0.174
0.35	399	-4.3	0.107
0.50	400	-9.5	0.0002
0.65	399	-9.3	0.0007
0.75	397	-14.1	$< 10^{-4}$
0.85	392	-11.7	0.0001
pooled	2774	-6.3	$< 10^{-4}$

Table 8: The length effect in isolation: `filler` against `none` on `node_degree`, `qwen3-1.7b-think`, over a wider density range than the main sweep. Δ in percentage points. Content-free text is harmless on sparse graphs and costs up to 14.1 points on dense ones — more than most primer *contents* gain anywhere in this study.

p	n	plain	think	Δ
0.10	1185	0.929	0.958	+2.9
0.20	1196	0.781	0.901	+12.0
0.35	1199	0.425	0.655	+22.9
0.50	1200	0.292	0.581	+28.8
0.65	1552	0.126	0.503	+37.7
0.75	1554	0.094	0.476	+38.2
0.85	1553	0.054	0.554	+50.0
pooled	9439	0.318	0.610	+29.2

Table 9: Chain-of-thought against greedy decoding on the identical `qwen3-1.7b` checkpoint and byte-identical prompts, `node_degree`, pooled over the `none`, `components`, `clustering` and `degree` conditions. Δ in percentage points; every level significant at $p < 10^{-3}$ under an exact McNemar test, the pooled comparison beyond the range of exact computation ($z = 46.4$).

long primer’s effect is length rather than content.

5.6 Chain-of-thought dominates every primer effect

Table 9 isolates the reasoning-mode effect from the primer channel: same checkpoint, same prompts, decoding mode alone. It is worth +29.2 points pooled and +50.0 points at $p = 0.85$ — an order of magnitude above the largest primer effect in Table 6 that is not answer retrieval, and consistent with Wei et al. (2022). Note also that it grows with density while the primer effects do not, which is what one expects of an intervention that adds computation rather than information.

The consequence for this study is visible in the last two blocks of Table 6: the thinking arms are at ceiling on `cycle_check`, `edge_existence` and `node_count` for every condition, and `qwen3-4b-think` is at ceiling on five of six

Sample	Δ	n	95% CI	p
<i>Original sample</i>				
$p = 0.10$	+0.3	400	—	1.000
$p = 0.20$	+3.3	400	—	0.182
$p = 0.35$	+7.5	400	—	0.009
$p = 0.50$	+4.3	400	—	0.155
pooled	+3.8	1600	[+1.5, +6.2]	0.0017
<i>Replication, independent graphs</i>				
$p = 0.10$	+2.0	400	—	0.185
$p = 0.20$	+4.0	400	—	0.101
$p = 0.35$	+4.5	400	—	0.145
$p = 0.50$	+6.5	400	—	0.028
pooled	+4.3	1600	[+1.9, +6.6]	0.0006

Table 10: `qwen3-1.7b`, `node_degree`, `clustering` against `none`, $n=40$, 400 paired instances per density level. Δ in percentage points; p from the exact McNemar test (permutation p pooled: 0.0015 and 0.0007). Per-level rows are descriptive and are a family of four — none survives correction on its own, which is the expected picture for an effect of this size at 400 pairs. The effect is positive at every level in both samples, which eight independent coin flips would produce with probability $1/256$.

tasks, leaving no headroom in which a primer effect could be observed. Where chain-of-thought does not saturate a task it censors it instead (Section 5.1). A primer study conducted only on reasoning-mode models of this scale would have almost no interpretable cells to report at all.

Primers do, however, change how often the thinking arms fail to terminate — an effect on the *process* rather than on accuracy, and one that would be invisible to an accuracy-only analysis. For `qwen3-1.7b-think`, `degree` truncates 3.4 points more often than `none` ($p < 0.001$) while all (-3.4), `filler` (-1.8) and `clustering` (-1.3) truncate less; for `qwen3-4b-think`, every primer truncates less than `none`, by 2.0 to 4.3 points (all $p < 0.001$). A primer therefore changes how long the model thinks even where it does not change what the model concludes.

5.7 The one cell that survives every control

One cell clears every control applied in this study, and we report it separately because the power available to it is four times that of any single cell in Table 6. On a dedicated sweep of 400 graphs per density level, the `clustering` primer improves `qwen3-1.7b` on `node_degree` by +3.8 points [+1.5, +6.2], and an independent sample at the same design points reproduces it at +4.3 points [+1.9, +6.6]; see Table 10.

The effect survives each control in turn. It

Primer	Δ vs none	
	dropped	zeroed
rwse	-7.5	-0.2
clustering	+0.3	+8.5
all	-1.1	+7.0

Table 11: Where the two conventions for truncated generations disagree, on `qwen3-1.7b / cycle_check`. Dropping the pair (this paper) and scoring it incorrect (`check_significance.py`) give opposite verdicts on 1.7B `cycle_check`, because `none` truncates on 9% of that task’s rows while `clustering` and `all` truncate on under 1%. Neither convention is uniformly more conservative; both are reported so the disagreement is visible rather than buried in a pipeline choice.

is not answer leakage: the shortcut ceiling for `clustering` on `node_degree` is 0.082 (Table 2), the same as `none`, because a clustering coefficient does not determine a degree. It is not length: `clustering` is 200 characters *shorter* than the null-content `filler` (Table 1), which registers no effect on this task in either 1.7B arm (-0.5 and -3.5 , both n.s.). And it is not a single lucky density level: it is positive at all four in both samples.

We are nonetheless cautious about it, for three reasons. The main sweep is consistent with it but cannot confirm it: at 100 graphs per cell it measures $+3.0$ points ($p = 0.256$), which is what a fourfold reduction in sample size predicts for an effect of this magnitude, not a failed replication. Under chain-of-thought the corresponding estimates are $+1.0$ ($p = 0.694$) in the main sweep and -1.1 ($p = 0.244$) on the wide-density sweep, so the effect does not survive into the thinking arm. And `qwen3-4b` cannot test it at all, being saturated under `none` (Table 7). We therefore report it as a single, replicated, model- and mode-specific effect, not as a general property of clustering primers.

For completeness: `components` was tested on the same original sweep and is inert pooled ($+0.1$ points, $p = 0.948$), but its paired difference trends negative with density (slope -0.148 per unit density, permutation $p = 0.022$), driven by -5.0 points at $p = 0.50$. We do not interpret this — it is one slope out of a small family — but record it because `components` is the one condition whose effects cannot be length.

5.8 Sensitivity to the truncation convention

Excluding truncated generations rather than scoring them incorrect is a choice, and on two tasks it changes the verdict rather than the third decimal place. Table 11 gives the case where it matters most. Under the drop convention `rwse` costs 1.7B 7.5 points on `cycle_check`; under the zero convention that effect vanishes and two others appear, because the arms being compared truncate at very different rates. The drop convention answers “given that both arms produced an answer, which was right more often?”; the zero convention answers “which arm produced a correct answer more often?”. Both are legitimate, and a primer that makes a model ramble until it runs out of budget has caused a real failure that the drop convention forgives. We use the drop convention throughout and report the disagreement here; on every task other than `cycle_check` and `edge_count` the two agree, because truncation is under 1%.

6 Discussion

The central negative result of this study is a failure to replicate across scale: of the 144 (arm, task, primer) comparisons in Table 6, no primer effect that clears both the `none` baseline and the `filler` control in one model size does so in the other. Three regularities account for most of what the table contains, and none of them is the mechanism a primer was hypothesized to supply.

First, a primer helps reliably only where it states the requested quantity verbatim, and even then only partially. `degree` and `all` move `node_degree` and `edge_count` and move nothing else consistently; the derived statistics that do not name an answer — `clustering`, `rwse`, `components` — produce no effect that survives both controls in more than one arm. This is answer retrieval, the behaviour the shortcut solver was built to expose, and the models perform it badly: never more than 28 points of a 100-point available gain on `edge_count`, and at 4B in the wrong direction entirely. The natural reading is that a primer does not replace the model’s own computation but competes with it, and that the model with the better internal computation has more to lose.

Second, added text is an active treatment regardless of its content. The length-matched `filler` significantly harms both model sizes on `edge_existence` and `connected_nodes` and significantly helps both on `node_count`, at

magnitudes (-15.8 to $+39.8$ points) exceeding nearly every content effect measured here. An experimental design that compares a primer only against `none` — the design a naive reading of the primer hypothesis invites — therefore estimates content and length confounded, with the length term frequently the larger of the two. We would go further: a primer paper reporting gains against a no-primer baseline alone should be assumed to be reporting a length effect until a placebo says otherwise.

Third, the headroom in which any of this could be measured is largely consumed by the decoding mode. Chain-of-thought is worth $+29.2$ points pooled on an identical checkpoint (Table 9), and its effect is to move most (task, arm) cells to ceiling: `qwen3-4b-think` exceeds 97% accuracy under `none` on five of six tasks, and on those five no primer differs from `none` by more than 1 point. Where chain-of-thought does not saturate a task it censors it instead: the 62–81% truncation rate on `edge_count` (Table 4) removes that task from analysis in both thinking arms. A primer study conducted only on reasoning-mode models of this scale would have almost no interpretable cells to report.

Taken together, these results argue that the interesting quantity is not whether a structural preamble helps, but where it *could* help. Of the 24 (arm, task) cells in Table 6, 10 are at or above 98% under `none` and therefore admit no upward effect, 2 are censored past interpretability, and 2 more (`node_count`, `cycle_check`) have effectively constant gold answers at this graph size and measure response formatting rather than graph reasoning. The cells that remain are few enough to enumerate, and they are the ones on which the replicated `clustering` effect of Table 10 was found. Screening for interpretable cells before measuring, rather than measuring everywhere and interpreting afterwards, is the design lesson we would carry forward.

7 Conclusion

We evaluated seven primer conditions across six GraphQA tasks, four densities and four model arms at $n=40$ — 67,200 generations — against two controls a primer study needs and rarely has: a graph-blind solver that bounds how much of an effect is answer retrieval, and a length-matched, content-free primer that bounds how much is prompt length.

No primer effect clears both controls in both model sizes. The effects that do appear are either retrieval of a stated answer, which the models exploit only partially and which reverses sign at 4B, or responses to added text irrespective of its content. One exception survives every control and an independent replication: `clustering` on `node_degree` for `qwen3-1.7b`, $+3.8$ and $+4.3$ points. We read the broader result not as evidence that structural preambles cannot help, but as evidence that most (model, task) pairs at this scale afford no interpretable test of whether they do.

Limitations

This sweep covers only two model sizes (1.7B and 4B parameters) from a single model family, so the findings may not generalize to substantially larger models or to other architectures. The direction of the `degree` effect reverses between the two sizes we do have, which is a reason to expect further reversals rather than a stable trend.

Several (primer condition, task) cells are known in advance to be shortcut-dominated at this graph size and density range (Section 4); results on those cells bound what can be concluded about genuine graph reasoning even though we report them.

Graphs are drawn from a single generator family (Erdős–Rényi); whether primer effects transfer to other graph generators, which [Fatemi et al. \(2023\)](#) show matters for encoding effects, remains untested here, and, as noted in Section 2, we found directly that varying the generator family itself is not a safe way to test this, since doing so confounds structure with task difficulty.

Compute and time constraints limited this sweep to a single node-count/density design rather than a broader sweep over graph sizes. All results are at $n=40$, and the shortcut ceilings were fit at $n \leq 19$, so the one number we carry across sizes is the one we flag as not transferring.

Finally, the length control is a bracket rather than an exact match: the `filler` condition is longer than `degree` and `clustering` and shorter than `rwse` and `all` (Table 1), so it bounds the length term without eliminating it. A design holding rendered length exactly constant while varying structure would isolate content more sharply than subtracting a bracketing control, and we regard that as the natural next step.

Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data, and no annotation labour: gold answers are computed programmatically from the graphs rather than collected from people. We see no direct risk of harm arising from the findings, which are a negative result about a prompting technique. The principal societal cost is the compute expended, reported in Section 4.

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A Effect sizes with confidence intervals, all cells

Paired effect sizes for every (arm, task, primer) cell: Δ in percentage points against none, a 95% paired-bootstrap interval, the permutation p -value, and Δ under the alternative convention that scores truncated generations incorrect rather than dropping the pair (Section 5.8). n is the number of pairs in which neither arm truncated.

Task	Primer	<i>n</i>	Δ	95% CI	perm <i>p</i>	zeroed
connected_nodes	components	400	+0.8	[-2.8, +4.5]	0.7861	+0.8
	clustering	400	+1.2	[-3.0, +5.8]	0.6458	+1.2
	rwse	400	+1.0	[-3.5, +5.2]	0.7304	+1.0
	degree	400	+3.5	[-0.5, +7.5]	0.1239	+3.5
	filler	400	-6.2	[-10.2, -2.2]	0.0040	-6.2
	all	399	+2.0	[-2.8, +6.8]	0.4644	+2.0
cycle_check	components	343	-0.6	[-1.5, +0.0]	0.5033	+2.0
	clustering	362	+0.3	[+0.0, +0.8]	1.0000	+8.5
	rwse	358	-7.5	[-10.3, -4.7]	0.0001	-0.2
	degree	301	-12.6	[-16.6, -9.0]	0.0001	-17.5
	filler	341	+0.3	[+0.0, +0.9]	1.0000	+3.0
	all	362	-1.1	[-2.5, +0.0]	0.2210	+7.0
edge_count	components	211	-1.4	[-3.8, +0.5]	0.3783	-1.0
	clustering	238	-0.4	[-2.5, +1.7]	1.0000	-0.5
	rwse	239	-0.8	[-2.9, +1.3]	0.6837	-0.8
	degree	253	+4.0	[+0.8, +7.1]	0.0179	+3.0
	filler	224	-0.4	[-2.2, +1.3]	1.0000	-0.5
	all	211	+2.4	[-0.5, +5.7]	0.2269	+0.8
edge_existence	components	400	-8.2	[-11.8, -4.8]	0.0001	-8.2
	clustering	400	+3.8	[-0.5, +8.0]	0.1199	+3.8
	rwse	399	+5.3	[+0.8, +9.8]	0.0321	+5.0
	degree	400	+4.0	[-0.5, +8.2]	0.0957	+4.0
	filler	400	-15.8	[-20.0, -11.8]	0.0001	-15.8
	all	400	+11.5	[+6.8, +16.2]	0.0001	+11.5
node_count	components	400	+21.2	[+15.8, +26.8]	0.0001	+21.2
	clustering	400	+67.2	[+62.7, +71.8]	0.0001	+67.2
	rwse	400	+2.0	[-4.2, +8.5]	0.5871	+2.0
	degree	400	+6.8	[+3.0, +10.5]	0.0008	+6.8
	filler	400	+39.8	[+35.0, +44.8]	0.0001	+39.8
	all	400	+47.0	[+42.0, +52.0]	0.0001	+47.0
node_degree	components	400	+1.8	[-2.0, +5.8]	0.4599	+1.8
	clustering	400	+3.0	[-1.8, +7.8]	0.2577	+3.0
	rwse	399	+4.0	[-0.5, +8.5]	0.1104	+4.0
	degree	400	+7.5	[+3.0, +12.0]	0.0018	+7.5
	filler	400	-0.5	[-4.5, +3.5]	0.9028	-0.5
	all	400	+10.2	[+5.2, +15.2]	0.0003	+10.2

Table 12: qwen3-1.7b: all cells.

Task	Primer	<i>n</i>	Δ	95% CI	perm <i>p</i>	zeroed
connected_nodes	components	400	+9.2	[+6.0, +12.8]	0.0001	+9.2
	clustering	400	+1.0	[-3.0, +4.8]	0.6893	+1.0
	rwse	400	+0.0	[-4.0, +4.0]	1.0000	+0.0
	degree	400	-3.0	[-7.2, +1.2]	0.1882	-3.0
	filler	400	-12.5	[-17.0, -8.0]	0.0001	-12.5
	all	400	-2.8	[-7.0, +1.5]	0.2511	-2.8
cycle_check	components	391	-1.3	[-2.6, +0.0]	0.1318	-2.0
	clustering	395	+0.3	[+0.0, +0.8]	1.0000	+0.5
	rwse	392	+0.3	[+0.0, +0.8]	1.0000	-0.2
	degree	391	-1.5	[-2.8, -0.5]	0.0312	-2.2
	filler	392	-0.3	[-0.8, +0.0]	1.0000	-0.5
	all	395	-0.5	[-1.5, +0.5]	0.6269	-0.2
edge_count	components	397	-1.3	[-2.8, +0.3]	0.1756	-1.2
	clustering	399	-1.3	[-3.0, +0.3]	0.2298	-1.2
	rwse	398	-1.5	[-3.0, -0.3]	0.0668	-1.5
	degree	393	+27.7	[+23.2, +32.6]	0.0001	+27.3
	filler	400	+1.2	[-0.5, +3.2]	0.3012	+1.2
	all	397	+3.8	[+1.3, +6.3]	0.0049	+3.8
edge_existence	components	400	-2.8	[-6.0, +0.5]	0.1254	-2.8
	clustering	400	+4.0	[+1.0, +7.0]	0.0168	+4.0
	rwse	400	-10.0	[-14.0, -6.2]	0.0001	-10.0
	degree	398	-4.3	[-7.8, -1.0]	0.0200	-4.8
	filler	400	-7.0	[-10.8, -3.5]	0.0002	-7.0
	all	399	-9.8	[-13.8, -6.0]	0.0001	-10.0
node_count	components	400	+9.0	[+6.2, +12.0]	0.0001	+9.0
	clustering	400	+8.8	[+5.8, +11.8]	0.0001	+8.8
	rwse	400	+9.2	[+6.2, +12.2]	0.0001	+9.2
	degree	400	+8.0	[+5.0, +11.2]	0.0001	+8.0
	filler	400	+9.2	[+6.5, +12.2]	0.0001	+9.2
	all	400	+9.8	[+7.0, +12.8]	0.0001	+9.8
node_degree	components	400	+0.2	[+0.0, +0.8]	1.0000	+0.2
	clustering	400	-0.5	[-2.0, +0.8]	0.7246	-0.5
	rwse	400	-2.2	[-4.0, -0.5]	0.0240	-2.2
	degree	400	-6.5	[-9.2, -3.8]	0.0001	-6.5
	filler	400	-0.5	[-2.0, +0.8]	0.7232	-0.5
	all	400	-9.2	[-12.2, -6.2]	0.0001	-9.2

Table 14: qwen3-4b: all cells.

Task	Primer	<i>n</i>	Δ	95% CI	perm <i>p</i>	zeroed
connected_nodes	components	392	+2.8	[-0.5, +6.1]	0.1268	+2.5
	clustering	394	+1.5	[-2.5, +5.6]	0.5384	+2.0
	rwse	393	-0.3	[-4.1, +3.6]	1.0000	+0.2
	degree	390	+7.4	[+3.6, +11.3]	0.0001	+6.2
	filler	394	+2.0	[-1.5, +5.6]	0.3275	+2.5
	all	392	+6.1	[+2.6, +9.9]	0.0027	+6.2
cycle_check	components	346	+0.0	[+0.0, +0.0]	1.0000	-7.5
	clustering	376	+0.0	[+0.0, +0.0]	1.0000	-0.5
	rwse	386	+0.0	[+0.0, +0.0]	1.0000	+2.2
	degree	295	-0.7	[-1.7, +0.0]	0.5036	-21.8
	filler	388	+0.0	[+0.0, +0.0]	1.0000	+3.0
	all	373	+0.0	[+0.0, +0.0]	1.0000	-0.8
edge_count	components	27	+7.4	[+0.0, +18.5]	0.5037	-0.8
	clustering	23	-8.7	[-26.1, +8.7]	0.6251	-1.2
	rwse	25	-4.0	[-12.0, +0.0]	1.0000	-3.2
	degree	13	-61.5	[-84.6, -38.5]	0.0102	-3.2
	filler	29	+3.4	[-6.9, +13.8]	1.0000	-1.5
	all	24	-8.3	[-29.2, +12.5]	0.6794	+5.0
edge_existence	components	398	-0.8	[-2.3, +0.8]	0.5090	-0.8
	clustering	398	-1.0	[-2.5, +0.3]	0.2911	-1.0
	rwse	398	-1.3	[-3.0, +0.3]	0.2274	-1.2
	degree	397	-2.3	[-4.3, -0.5]	0.0249	-2.5
	filler	399	-0.8	[-2.3, +0.8]	0.5051	-0.5
	all	399	-2.0	[-3.8, -0.5]	0.0374	-1.8
node_count	components	342	+0.9	[-0.3, +2.3]	0.3758	+1.5
	clustering	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	rwse	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	degree	360	+0.8	[-0.6, +2.2]	0.4512	+5.5
	filler	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	all	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
node_degree	components	395	-1.0	[-4.3, +2.3]	0.6543	-1.0
	clustering	396	+1.0	[-2.8, +4.8]	0.7010	+1.5
	rwse	394	+2.5	[-1.5, +6.9]	0.2898	+2.5
	degree	376	+11.7	[+7.4, +16.0]	0.0001	+8.8
	filler	396	-3.5	[-7.3, +0.3]	0.0954	-3.0
	all	396	+11.6	[+7.3, +15.9]	0.0001	+12.2

Table 13: qwen3-1.7b-think: all cells.

Task	Primer	<i>n</i>	Δ	95% CI	perm <i>p</i>	zeroed
connected_nodes	components	377	+0.5	[-0.5, +1.6]	0.6249	+0.5
	clustering	373	+0.0	[-1.3, +1.3]	1.0000	-0.8
	rwse	379	+0.0	[-1.3, +1.6]	1.0000	+0.5
	degree	376	+0.0	[-1.6, +1.6]	1.0000	+0.5
	filler	379	+0.8	[-0.3, +2.1]	0.3787	+1.2
	all	375	-0.5	[-2.1, +0.8]	0.7306	-1.2
cycle_check	components	381	+0.0	[+0.0, +0.0]	1.0000	-3.2
	clustering	394	+0.0	[+0.0, +0.0]	1.0000	+0.0
	rwse	395	+0.0	[+0.0, +0.0]	1.0000	+0.2
	degree	375	-0.8	[-1.9, +0.0]	0.2514	-5.5
	filler	396	+0.0	[+0.0, +0.0]	1.0000	+0.5
	all	392	+0.0	[+0.0, +0.0]	1.0000	-0.5
edge_count	components	60	+1.7	[+0.0, +5.0]	1.0000	+24.8
	clustering	59	+1.7	[-3.4, +6.8]	1.0000	+24.8
	rwse	49	+0.0	[-6.1, +6.1]	1.0000	+10.2
	degree	34	-23.5	[-38.2, -8.8]	0.0073	+8.5
	filler	50	-2.0	[-6.0, +0.0]	1.0000	+12.2
	all	41	-41.5	[-56.1, -26.8]	0.0002	+1.8
edge_existence	components	396	+0.0	[+0.0, +0.0]	1.0000	-0.5
	clustering	396	+0.0	[+0.0, +0.0]	1.0000	-0.8
	rwse	393	+0.0	[+0.0, +0.0]	1.0000	-1.2
	degree	398	+0.0	[+0.0, +0.0]	1.0000	+0.0
	filler	399	+0.0	[+0.0, +0.0]	1.0000	+0.0
	all	394	+0.0	[+0.0, +0.0]	1.0000	-1.0
node_count	components	400	+0.0	[-1.0, +1.0]	1.0000	+0.0
	clustering	399	+0.5	[+0.0, +1.3]	0.4936	+0.2
	rwse	399	+0.5	[+0.0, +1.3]	0.5005	+0.2
	degree	399	+0.5	[+0.0, +1.3]	0.5005	+0.2
	filler	400	+0.5	[+0.0, +1.2]	0.4870	+0.5
	all	400	+0.5	[+0.0, +1.2]	0.4870	+0.5
node_degree	components	380	-0.3	[-0.8, +0.0]	1.0000	+0.2
	clustering	386	-0.3	[-1.0, +0.5]	1.0000	+1.8
	rwse	381	-2.1	[-3.7, -0.8]	0.0072	-1.2
	degree	388	+0.0	[-0.8, +0.8]	1.0000	+2.5
	filler	386	+0.0	[+0.0, +0.0]	1.0000	+2.0
	all	383	-0.8	[-1.8, +0.0]	0.2466	+0.5

Table 15: qwen3-4b-think: all cells.

B Per-density accuracy, all arms and tasks

Exact-match accuracy for every (arm, task, density, primer) cell, with the per-cell count of generations dropped to truncation where any occurred. These are the numbers Table 6 pools over density.

B.1 qwen3-1.7b

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.920	0.920	0.900	0.980	0.970	0.920	0.909
0.2	0.910	0.900	0.900	0.860	0.920	0.860	0.840
0.35	0.590	0.640	0.630	0.610	0.640	0.490	0.740
0.5	0.480	0.470	0.520	0.490	0.510	0.380	0.500
<i>truncated rows dropped</i>							
0.1	0	0	0	0	0	0	1
0.2	0	0	0	0	0	0	0
0.35	0	0	0	0	0	0	0
0.5	0	0	0	0	0	0	0

Table 16: qwen3-1.7b, `connected_nodes`: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.988	0.978	1.000	0.946	0.989	1.000	0.969
0.2	1.000	1.000	1.000	0.940	0.857	1.000	1.000
0.35	1.000	0.989	1.000	0.870	0.851	1.000	0.990
0.5	1.000	1.000	1.000	0.930	0.831	1.000	0.990
<i>truncated rows dropped</i>							
0.1	17	10	2	7	11	5	3
0.2	9	9	1	0	9	10	0
0.35	9	7	0	0	26	9	1
0.5	1	0	0	0	23	1	0

Table 17: qwen3-1.7b, `cycle_check`: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.043	0.000	0.000	0.000	0.067	0.000	0.000
0.2	0.022	0.013	0.031	0.022	0.032	0.033	0.011
0.35	0.000	0.000	0.000	0.000	0.010	0.000	0.060
0.5	0.000	0.000	0.000	0.000	0.087	0.000	0.054
<i>truncated rows dropped</i>							
0.1	31	20	10	10	10	15	26
0.2	9	20	3	9	6	9	9
0.35	50	33	26	21	2	40	33
0.5	29	36	33	32	20	41	44

Table 18: qwen3-1.7b, `edge_count`: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.890	0.860	0.840	0.919	0.830	0.840	0.890
0.2	0.730	0.620	0.740	0.680	0.700	0.530	0.810
0.35	0.660	0.500	0.720	0.760	0.670	0.340	0.780
0.5	0.540	0.510	0.670	0.670	0.780	0.480	0.800
<i>truncated rows dropped</i>							
0.1	0	0	0	1	0	0	0
0.2	0	0	0	0	0	0	0
0.35	0	0	0	0	0	0	0
0.5	0	0	0	0	0	0	0

Table 19: qwen3-1.7b, `edge_existence`: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.020	0.190	0.950	0.320	0.030	0.160	0.700
0.2	0.000	0.410	1.000	0.170	0.050	0.700	0.520
0.35	0.280	0.700	1.000	0.550	0.450	0.990	0.940
0.5	0.960	0.810	1.000	0.300	1.000	1.000	0.980

Table 20: qwen3-1.7b, `node_count`: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.910	0.920	0.930	0.940	0.940	0.960	0.970
0.2	0.800	0.820	0.820	0.840	0.900	0.730	0.900
0.35	0.410	0.440	0.470	0.475	0.590	0.410	0.580
0.5	0.300	0.310	0.320	0.330	0.290	0.300	0.380
<i>truncated rows dropped</i>							
0.1	0	0	0	0	0	0	0
0.2	0	0	0	0	0	0	0
0.35	0	0	0	1	0	0	0
0.5	0	0	0	0	0	0	0

Table 21: qwen3-1.7b, `node_degree`: exact-match accuracy by density and primer, truncated rows excluded.

B.2 qwen3-1.7b-think

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.948	0.959	0.949	0.980	0.939	0.960	1.000
0.2	0.960	0.949	0.960	0.880	0.948	0.980	0.959
0.35	0.694	0.710	0.660	0.720	0.810	0.720	0.780
0.5	0.590	0.670	0.670	0.592	0.758	0.610	0.687
<i>truncated rows dropped</i>							
0.1	3	3	1	0	2	1	0
0.2	0	1	0	0	3	1	2
0.35	2	0	0	0	0	0	0
0.5	0	0	0	2	1	0	1

Table 22: qwen3-1.7b-think, `connected_nodes`: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.2	1.000	1.000	1.000	1.000	0.986	1.000	1.000
0.35	1.000	1.000	1.000	1.000	0.985	1.000	1.000
0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000
<i>truncated rows dropped</i>							
0.1	1	14	1	0	12	0	5
0.2	2	4	0	1	27	0	4
0.35	5	13	7	0	34	0	2
0.5	4	11	6	2	24	0	4

Table 23: qwen3-1.7b-think, cycle_check: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.960	0.977	0.912	0.975	0.550	0.929	0.800
0.2	0.652	0.630	0.781	0.846	0.696	0.667	0.682
0.35	–	–	0.500	–	0.765	1.000	0.667
0.5	–	–	–	–	0.625	–	0.595
<i>truncated rows dropped</i>							
0.1	50	56	66	60	80	58	70
0.2	77	73	68	87	77	76	78
0.35	100	100	96	100	83	98	67
0.5	100	100	100	100	84	100	63

Table 24: qwen3-1.7b-think, edge_count: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	1.000	0.990	0.980	1.000	0.970	0.970	0.960
0.2	1.000	0.970	0.980	0.950	0.950	0.980	0.960
0.35	0.970	0.980	0.990	0.970	0.970	1.000	0.970
0.5	1.000	1.000	0.980	1.000	0.990	0.990	1.000
<i>truncated rows dropped</i>							
0.1	1	1	1	0	1	0	0
0.2	0	0	0	0	0	0	0
0.35	0	0	0	0	0	0	0
0.5	0	0	0	1	1	0	0

Table 25: qwen3-1.7b-think, edge_existence: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	1.000	1.000	1.000	1.000	0.990	1.000	1.000
0.2	1.000	0.989	1.000	1.000	1.000	1.000	1.000
0.35	1.000	1.000	1.000	1.000	0.979	1.000	1.000
0.5	0.945	1.000	1.000	1.000	1.000	1.000	1.000
<i>truncated rows dropped</i>							
0.1	2	4	0	0	1	0	0
0.2	3	6	0	0	1	0	0
0.35	17	5	0	0	5	0	0
0.5	9	14	0	0	4	0	0

Table 26: qwen3-1.7b-think, node_count: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.928	0.939	0.949	0.960	0.980	0.980	0.980
0.2	0.950	0.890	0.970	0.929	0.939	0.850	0.970
0.35	0.600	0.620	0.610	0.650	0.837	0.560	0.820
0.5	0.600	0.580	0.590	0.636	0.820	0.550	0.780
<i>truncated rows dropped</i>							
0.1	3	2	1	0	0	1	0
0.2	0	0	0	2	2	0	1
0.35	0	0	0	0	8	0	0
0.5	0	0	0	1	11	0	0

Table 27: qwen3-1.7b-think, node_degree: exact-match accuracy by density and primer, truncated rows excluded.

B.3 qwen3-4b

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.930	1.000	0.850	0.830	0.860	0.720	0.820
0.2	0.880	0.990	0.970	0.830	0.900	0.740	0.880
0.35	0.840	0.940	0.890	0.890	0.800	0.780	0.870
0.5	0.790	0.880	0.770	0.890	0.760	0.700	0.760

Table 28: qwen3-4b, connected_nodes: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.990	0.969	1.000	1.000	0.990	0.990	1.000
0.2	1.000	0.990	1.000	1.000	0.990	1.000	1.000
0.35	1.000	1.000	1.000	1.000	0.969	1.000	1.000
0.5	1.000	0.980	1.000	1.000	0.980	1.000	0.970
<i>truncated rows dropped</i>							
0.1	0	3	2	4	3	4	2
0.2	3	3	0	1	0	0	0
0.35	0	0	0	0	2	1	0
0.5	0	0	0	0	1	0	0

Table 29: qwen3-4b, cycle_check: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.050	0.000	0.000	0.010	0.278	0.100	0.030
0.2	0.020	0.010	0.010	0.010	0.323	0.020	0.030
0.35	0.010	0.020	0.020	0.000	0.253	0.010	0.081
0.5	0.000	0.000	0.000	0.000	0.337	0.000	0.092
<i>truncated rows dropped</i>							
0.1	0	0	1	2	3	0	0
0.2	0	0	0	0	1	0	0
0.35	0	1	0	0	1	0	1
0.5	0	2	0	0	2	0	2

Table 30: qwen3-4b, edge_count: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.960	0.990	0.970	0.960	0.990	0.960	0.960
0.2	0.930	0.940	0.980	0.890	0.919	0.840	0.840
0.35	0.830	0.750	0.890	0.700	0.790	0.750	0.700
0.5	0.900	0.830	0.940	0.670	0.747	0.790	0.730
<i>truncated rows dropped</i>							
0.1	0	0	0	0	0	0	1
0.2	0	0	0	0	1	0	0
0.35	0	0	0	0	0	0	0
0.5	0	0	0	0	1	0	0

Table 31: qwen3-4b, edge_existence: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.920	0.970	0.970	0.980	0.940	0.980	1.000
0.2	0.690	1.000	0.990	1.000	0.990	1.000	1.000
0.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Table 32: qwen3-4b, node_count: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	1.000	1.000	1.000	1.000	0.940	1.000	0.880
0.2	0.990	0.990	1.000	0.970	0.980	0.990	0.960
0.35	0.990	0.990	0.990	0.960	0.920	0.990	0.920
0.5	0.990	1.000	0.960	0.950	0.870	0.970	0.840

Table 33: qwen3-4b, node_degree: exact-match accuracy by density and primer, truncated rows excluded.

B.4 qwen3-4b-think

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.979	0.989	1.000	1.000	1.000	1.000	0.978
0.2	0.989	1.000	0.989	1.000	0.979	0.989	0.978
0.35	1.000	1.000	1.000	0.970	1.000	0.990	0.990
0.5	0.980	0.990	0.980	0.990	0.980	1.000	0.990
<i>truncated rows dropped</i>							
0.1	3	6	10	4	6	2	9
0.2	5	6	7	8	3	6	7
0.35	3	1	1	0	2	0	0
0.5	2	1	0	0	1	3	1

Table 34: qwen3-4b-think, connected_nodes: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	1.000	1.000	1.000	1.000	0.979	1.000	1.000
0.2	1.000	1.000	1.000	1.000	0.990	1.000	1.000
0.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000
<i>truncated rows dropped</i>							
0.1	2	1	2	2	4	1	0
0.2	0	0	1	0	3	0	1
0.35	1	7	0	0	7	0	2
0.5	0	8	0	0	8	0	2

Table 35: qwen3-4b-think, cycle_check: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.975	0.986	0.975	0.913	0.778	1.000	0.680
0.2	1.000	0.970	0.981	0.923	0.750	0.920	0.528
0.35	0.875	1.000	0.918	1.000	0.634	0.957	0.378
0.5	–	1.000	1.000	1.000	0.600	1.000	0.297
<i>truncated rows dropped</i>							
0.1	60	27	20	31	64	40	50
0.2	66	34	47	48	52	50	47
0.35	92	65	51	91	59	77	63
0.5	100	92	96	99	60	99	63

Table 36: qwen3-4b-think, edge_count: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.2	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000
<i>truncated rows dropped</i>							
0.1	1	1	3	4	1	1	4
0.2	0	1	1	2	0	0	1
0.35	0	0	0	0	0	0	0
0.5	0	1	0	0	0	0	0

Table 37: qwen3-4b-think, edge_existence: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.980	0.990	1.000	1.000	1.000	1.000	1.000
0.2	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.35	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.5	1.000	0.990	1.000	1.000	1.000	1.000	1.000
<i>truncated rows dropped</i>							
0.1	0	0	1	1	0	0	0
0.2	0	0	0	0	0	0	0
0.35	0	0	0	0	0	0	0
0.5	0	0	0	0	1	0	0

Table 38: qwen3-4b-think, node_count: exact-match accuracy by density and primer, truncated rows excluded.

p	none	compon	cluste	rwse	degree	filler	all
0.1	0.990	1.000	1.000	1.000	1.000	1.000	1.000
0.2	1.000	0.990	0.990	0.980	0.990	1.000	0.990
0.35	1.000	1.000	1.000	1.000	1.000	1.000	0.980
0.5	1.000	0.990	0.990	0.940	1.000	1.000	1.000
<i>truncated rows dropped</i>							
0.1	3	6	3	7	1	3	5
0.2	5	1	0	2	0	0	1
0.35	3	2	0	0	0	0	0
0.5	0	0	0	0	0	1	1

Table 39: qwen3-4b-think, node_degree: exact-match accuracy by density and primer, truncated rows excluded.